

# AMERI MAJD

## Software Engineering Student

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## PROFESSIONAL SUMMARY

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I am a motivated Software Engineering student specializing in Machine Learning and Data Science. Possesses hands-on experience in developing end-to-end AI systems, from data preprocessing and model training to deployment and optimization for embedded and web platforms. Eager to apply a strong technical skill set to challenging roles in AI, ML, and Software Engineering

## WORK EXPERIENCE

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### AI / Machine Learning Engineering Intern

ACTIA Engineering Services – Tunis, Tunisia

July 2025 – September 2025

- Developed and optimized a machine learning model for real-time classification of pedestrians and cyclists using mmWave radar data on a Texas Instruments IWR6843AOP embedded platform.
- Engineered a complete data preprocessing pipeline in Python for the RadarScenes dataset (4M+ detections), handling NaN values, normalizing features, and generating fixed-size point clouds for model input.
- Benchmarked multiple architectures (PointNet, DGCNN, MLP) achieving up to 93.5% accuracy in distinguishing between vulnerable road users.
- Overcame critical embedded systems constraints (<1.4MB SRAM, <50ms inference time) via architectural pruning and post-training quantization, reducing model size by 99% and latency to <0.15ms while maintaining high performance.
- Implemented radar-to-radar data fusion concepts to enhance detection robustness and classification accuracy.

### Data Science Intern

HoutaLik – Monastir, Tunisia

June 2025 – July 2025

- Developed an end-to-end AI-powered system for automated resume classification and scoring, streamlining the recruitment process.
- Implemented web scraping with BeautifulSoup to build a diverse training dataset of job descriptions.
- Engineered a robust NLP pipeline using TF-IDF and spaCy for text preprocessing and feature extraction.
- Trained and evaluated multiple machine learning models (Logistic Regression, SVM, Random Forest), achieving high accuracy in categorizing resumes and matching them to relevant job postings.
- Built an interactive web application, allowing users to upload resumes and receive instant compatibility scores with different job roles.

## EDUCATION

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### Bachelor's degree in computer science

Higher Institute of Computer Science and Mathematics of Monastir – Monastir, Tunisia

Current

## PROJECTS & HACKATHONS

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### DataInsight – Data Extraction & Analysis Platform

May 2025

built a full-stack application to automate data collection and analysis, featuring high-performance scraping pipelines developed with FastAPI (Python) to extract data from web pages. It includes an interactive React.js interface for data exploration, designed workflows to clean, structure, and visualize datasets, and the capability to transform raw, unstructured web data into decision-ready information. Additionally, the platform allows users to download the processed data in various formats, including PDF, Word, or CSV, demonstrating strong skills in data engineering, preprocessing, and analysis while providing a versatile tool for data management and utilization.

### House Price Prediction with XGBoost

Decembre 2024

optimized an XGBoost regression model through hyperparameter tuning with GridSearchCV, improving prediction accuracy (RMSE) by 15% over baseline models. The process included preprocessing data by handling missing values and feature scaling, followed by visualizing key drivers of property value to provide actionable insights. This project showcases a strong ability to enhance model performance and interpret complex datasets effectively.

### Xtreme V18.0 Hackathon

2024

Ranked 12th nationally in Tunisia. Implemented real-time problem-solving and technical troubleshooting.

### MAE V2.0 Hackathon

2025

Selected among top 103 participants from 1,400+ applicants; Developed an idea of a smart insurance claim system.

### Xtreme V19.0 Hackathon

2024

Ranked 2<sup>nd</sup> nationally in Tunisia . Implemented real-time problem-solving and technical troubleshooting.

## SKILLS

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- **Programming Languages:** Python / SQL / C, C++, Java
- **Technical Knowledge:** Machine Learning / Embedded AI / Competitive Programming / Deep Learning / NLP / Data Preprocessing

## LANGUAGE SKILLS

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- **Mother Tongue:** Arabic
- **Other Languages:** English(B2), French(B2), German(A2)